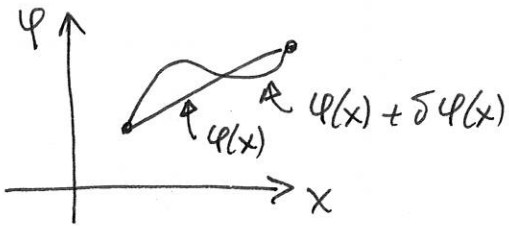
Note on Functional derivative

Let $F[\varphi] = \int_a^b dx\, K[\varphi(x)]$ be a functional so that it is a mapping from a linear space of functions $\{\varphi(x)\}$ to the field of real (or complex) numbers F.



The object $\frac{\delta F[\varphi]}{\delta \varphi(x)}$ tells how the value $F[\varphi]$ changes when function $\varphi(x)$ is changed at point x

$$\delta F[\varphi] = \int_a^b \frac{\delta F[\varphi]}{\delta \varphi(x)}\, \delta\varphi(x)\, dx$$

Recall that in case of the action $S = \int dx\, L(\varphi, \varphi')$

$$\frac{\delta S}{\delta \varphi} = \frac{\partial L}{\partial \varphi} - \frac{d}{dx}\frac{\partial L}{\partial \varphi'}$$

gives us the right hand-side of the Euler-Lagrange equation (EL)

We are free to choose any $\delta\varphi(x) \Rightarrow$ let $\delta\varphi(x) = \varepsilon\,\delta(x-y)$ be a variation at a single point $y \in (a,b)$, of strength ε.

Then $\delta F[\varphi] = F[\varphi + \varepsilon\delta(x-y)] - F[\varphi] = \int dx\, \frac{\delta F[\varphi]}{\delta\varphi(x)}\, \varepsilon\,\delta(x-y) =$

$$= \varepsilon\, \frac{\delta F}{\delta \varphi(y)} \Rightarrow \frac{\delta F[\varphi]}{\delta\varphi(y)} = \lim_{\varepsilon\to 0} \frac{F[\varphi + \varepsilon\delta(x-y)] - F[\varphi]}{\varepsilon}$$

1) So let $F[\varphi] = \int dx \, (\varphi(x))^n$

$$\frac{\delta F[\varphi]}{\delta \varphi(y)} = \lim_{\varepsilon \to +0} \frac{1}{\varepsilon} \int dx \left\{ \left(\varphi(x) + \varepsilon \delta(x-y)\right)^n - \left(\varphi(x)\right)^n \right\} =$$

$$= \lim_{\varepsilon \to 0} \frac{1}{\varepsilon} \int dx \; n \varepsilon \left(\varphi(x)\right)^{n-1} \delta(x-y) = n \left(\varphi(y)\right)^{n-1}.$$

Hence $\frac{\delta}{\delta \varphi(y)} \int dx \; K[\varphi(x)] = K'[\varphi(y)]$.

2) $F[\varphi] = \int dx \left(\frac{d\varphi(x)}{dx}\right)^n$

$$\frac{\delta F[\varphi]}{\delta \varphi(y)} = \lim_{\varepsilon \to 0} \frac{1}{\varepsilon} \int dx \left(\frac{d}{dx}\left[\varphi(x) + \varepsilon \delta(x-y)\right]\right)^n - \left(\frac{d}{dx}\varphi(x)\right)^n$$

$$= \lim_{\varepsilon \to 0} \frac{1}{\varepsilon} \int dx \left\{ \left(\frac{d\varphi}{dx} + \varepsilon \frac{d}{dx} \delta(x-y)\right)^n - \left(\frac{d\varphi}{dx}\right)^n \right\}$$

$$= \lim_{\varepsilon \to 0} \frac{1}{\varepsilon} \int dx \left\{ n \left(\frac{d\varphi(x)}{dx}\right)^{n-1} \varepsilon \frac{d}{dx} \delta(x-y) + O(\varepsilon^2) \right\}$$

$$= n \int dx \left(\frac{d\varphi}{dx}\right)^{n-1} \frac{d}{dx} \delta(x-y) = -n \frac{d}{dx} \left(\frac{d\varphi}{dx}\right)^{n-1} \Big|_{x=y}.$$

Thus for $H = \int dx \left\{ \frac{p^2(x)}{2\rho} + \frac{1}{2} K_0 \left(\frac{\partial u(x)}{\partial x} \right)^2 \right\}$

$$\frac{\delta H}{\delta p(y)} = \frac{1}{2\rho} \cdot 2\, p(y)$$

$$\frac{\delta H}{\delta u(y)} = \left(\frac{1}{2} K_0\right)(-2) \frac{d}{dx} \left(\frac{\partial u(x)}{\partial x} \right)^{2-1} \Bigg|_{x=y} =$$

$$= -K_0 \frac{\partial^2 u}{\partial y^2} .$$

Hence
$$\begin{cases} \dfrac{\partial u(x)}{\partial t} = \dfrac{\delta H}{\delta p(x)} = \dfrac{p(x)}{\rho} \\[2ex] \dfrac{\partial p(x)}{\partial t} = -\dfrac{\delta H}{\delta u(x)} = +K_0 \dfrac{\partial^2 u}{\partial x^2} \end{cases}$$

$$u(x,t) = \sum_k u_k e^{i(kx-\omega_k t)} + u_k^* e^{-i(kx-\omega_k t)} \quad , \omega_k = \sqrt{\frac{K_0}{\rho}}\, k$$

$$H = \frac{1}{2}\int dx \left\{ \frac{p^2}{\rho} + K_0 \left(\frac{\partial u}{\partial x}\right)^2 \right\}$$

$$p = \rho\, \dot{u} = \rho \sum_k \omega_k \left(-i u_k e^{i(kx-\omega t)} + i u_k^* e^{-i(kx-\omega_k t)}\right)$$

$$u' = \sum_k ik \left(u_k e^{i(kx-\omega_k t)} - u_k^* e^{-i(kx-\omega_k t)}\right)$$

$$H = \frac{1}{2}\int dx \sum_{kk'} (-i)^2 \rho\, \omega_k \omega_{k'} \left(u_k e^{ikx-\omega_k t} - u_k^* e^{-i(kx-\omega_k t)}\right)$$
$$\left(u_{k'} e^{i(k'x-\omega_k' t)} - u_{k'}^* e^{-i(k'x-\omega_k' t)}\right)$$
$$+ K_0 i^2 kk'$$

$$= -\frac{1}{2}\int dx \sum_{kk'} \left(\rho\omega_k\omega_{k'} + K_0 kk'\right)\Big(u_k u_{k'} e^{i(k+k')x - i(\omega+\omega')t}$$
$$+ u_k^* u_{k'}^* e^{-i(k+k')x + i(\omega+\omega')t} - u_k^* u_{k'} e^{-i(k-k')x + i(\omega-\omega')t}$$
$$- u_k u_{k'}^* e^{i(k-k')x - i(\omega-\omega')t}\Big)$$

$\Rightarrow$ for $k = k'$: $\rho\,\omega_k\omega_{k'} + K_0 kk' = \rho\,\omega_k^2 + K_0 k^2 = 2\rho\,\omega_k^2$

$k = -k'$: $\rho\,\omega_k^2 - K_0 k^2 = 0$

$$\Rightarrow H = \frac{1}{2}\sum_k 2\rho\,\omega_k^2 V (u_k^* u_k + u_k u_k^*)$$

$$\int dx\, e^{i(k-k')x} = V \delta_{k,k'}$$

$\Rightarrow$ Define $d_k = \sqrt{4\rho V \omega_k}\, u_k$: $H = \sum_k \omega_k d_k^* d_k$.

$$Q_k = \frac{1}{\sqrt{2\omega_k}}(d_k + d_k^*), \quad P_k = -i\sqrt{\frac{\omega_k}{2}}(d_k - d_k^*)$$

$$\sqrt{2\omega_k}\, Q_k \pm i\sqrt{\frac{2}{\omega_k}}\, P_k = \begin{cases} 2d_k \\ 2d_k^* \end{cases} \Rightarrow d_k = \frac{\omega_k Q_k + iP_k}{\sqrt{2\omega_k}}$$

$$d^* = \frac{\omega_k Q_k - iP_k}{\sqrt{2\omega_k}}$$

$$H = \sum_k \omega_k \frac{\omega_k Q_k - iP_k}{\sqrt{2\omega_k}} \frac{\omega Q + iP_k}{\sqrt{2\omega_k}} = \sum_k \frac{1}{2}\left(P_k^2 + \omega_k^2 Q_k^2\right)$$

Mode decomposition again:

We found $H = \sum_k H_k = \sum_k \frac{1}{2}(P_k^2 + \omega_k^2 Q_k^2)$

where $\begin{cases} Q_k = \dfrac{d_k + d_k^*}{\sqrt{2\omega_k}} = \sqrt{\dfrac{2\rho L \omega_k}{2\omega_k}}\,(u_k + u_k^*) \\ P_k = -i\sqrt{\dfrac{\omega_k}{2}}\,\sqrt{2\rho L \omega_k}\,(u_k - u_k^*) \end{cases}$

are simply <u>Re</u> and <u>Im</u> parts of time-independent complex amplitude u_k.

But recall that $u(x,t) = \sum_k \left(u_k e^{i(kx - \omega_k t)} + u_k^* e^{-i(kx - \omega_k t)}\right)$

u_k always appears in combination with $e^{-i\omega_k t}$

$\Rightarrow$ introduce t-dependent amplitude $u_k(t) = u_k e^{-i\omega_k t}$.

$\Rightarrow u_k^*(t) = u_k^* e^{i\omega_k t}$.

$$\Rightarrow \begin{cases} Q_k(t) = \sqrt{\rho L}\,(u_k(t) + u_k^*(t)) \\ P_k(t) = (-i)\sqrt{\rho L}\,\omega_k\,(u_k(t) - u_k^*(t)) \end{cases}$$

Observe now that $\dot{Q}_k(t) = \dfrac{\partial Q_k}{\partial t} = \sqrt{\rho L}\,(-i\omega_k u_k(t) + i\omega_k u_k^*(t)) = (-i)\underbrace{\omega_k \sqrt{\rho L}\,(u_k(t) - u_k^*(t))}_{P_k(t)/(-i)} =$

$= P_k(t)$

$$\boxed{\begin{aligned} &\text{here we use} \\ &\dot{u}_k(t) = -i\omega_k u_k(t) \\ &\dot{u}_k^*(t) = i\omega_k u_k^*(t) \end{aligned}}$$

$$\dot{Q}_k(t) = P_k(t)$$

Also, $\dot{P}_k(t) = \sqrt{\rho L}\,(-i\omega_k)\,(-i\omega_k u_k(t) - i\omega_k u_k^*(t))$

$= \sqrt{\rho L}\,(-i\omega_k)^2 (u_k(t) + u_k^*(t)) = -\omega_k^2 Q_k(t)$

Thus equations of motion for $Q_k(t), P_k(t)$

$$\begin{cases} \dot{Q}_k(t) = P_k(t) \\ \dot{P}_k(t) = -\omega_k^2 Q_k(t) \end{cases}$$

coincide with those of canonically-conjugated pair (Q_k, P_k) describing oscillator $H_k = \frac{1}{2}(P_k^2 + \omega_k^2 Q_k^2)$

$$\begin{cases} \dot{Q}_k = \dfrac{\partial H}{\partial P_k} = P_k \\ \dot{P}_k = -\dfrac{\partial H}{\partial Q_k} = -\omega_k^2 Q_k \end{cases}$$

(from Hamilton's eqns)

Now <u>quantize</u> $H_k \to \hat{H}_k = \frac{1}{2}(P_k^2 + \omega_k^2 \hat{Q}_k^2)$ Single oscillator

where $[\hat{Q}_k, \hat{P}_k] = i\hbar$

Introduce $\begin{cases} \hat{a}_k = \dfrac{\omega_k \hat{Q}_k + i\hat{P}_k}{\sqrt{2\hbar\omega_k}} \\ \hat{a}_k^+ = \dfrac{\omega_k \hat{Q}_k - i\hat{P}_k}{\sqrt{2\hbar\omega_k}} \end{cases} \Rightarrow \begin{cases} \hat{Q}_k = \sqrt{\dfrac{\hbar}{2\omega_k}}(\hat{a}_k + \hat{a}_k^+) \\ \hat{P}_k = -i\sqrt{\dfrac{\hbar\omega_k}{2}}(\hat{a}_k - \hat{a}_k^+) \end{cases}$

so that $[\hat{a}_k, \hat{a}_k^+] = \dfrac{1}{2\hbar\omega_k}\Big(\omega_k^2 [\hat{Q}_k, \hat{Q}_k] + [\hat{P}_k, \hat{P}_k]$

$+ i\omega_k [\hat{P}_k, \hat{Q}_k] - i\omega_k [\hat{Q}_k, \hat{P}_k] \Big) =$

$= \dfrac{(-2i\omega_k)[\hat{Q}_k, \hat{P}_k]}{2\hbar\omega_k} = 1.$

$\hat{a}_k \hat{a}_k^+ - \hat{a}_k^+ \hat{a}_k = 1.$

As a result, quantum Hamiltonian reads

$$\hat{H}_k = \frac{1}{2}\left\{ -\frac{\hbar\omega_k}{2}\left(\hat{a}_k - \hat{a}_k^+\right)\left(\hat{a}_k - \hat{a}_k^+\right) + \omega_k^2 \frac{\hbar}{2\omega_k}\left(\hat{a}_k + \hat{a}_k^+\right)\left(\hat{a}_k + \hat{a}_k^+\right)\right\}$$

$$= \frac{1}{4}\hbar\omega_k \left\{ -\left(\hat{a}_k\hat{a}_k - \hat{a}_k\hat{a}_k^+ - \hat{a}_k^+\hat{a}_k + \hat{a}_k^+\hat{a}_k^+\right) + \right.$$

$$\left. + \hat{a}_k\hat{a}_k + \hat{a}_k\hat{a}_k^+ + \hat{a}_k^+\hat{a}_k + \hat{a}_k^+\hat{a}_k^+ \right\}$$

$$= \frac{1}{2}\hbar\omega_k \left(\underbrace{\hat{a}_k\hat{a}_k^+}_{\Downarrow \atop 1 + \hat{a}_k^+\hat{a}_k} + \hat{a}_k^+\hat{a}_k\right) = \hbar\omega_k\left(\hat{a}_k^+\hat{a}_k + \frac{1}{2}\right).$$